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ANTENNA ARRAY AND GROUND STATION

Abstract

An antenna array and a ground station, including at least one radiation module and a feeding module, the radiation module includes a radiation layer and a dielectric layer arranged in stack, the radiation layer includes a plurality of radiators, each radiator radiates signals; the feeding module includes a coupling layer arranged in stack, the feeding layer includes a plurality of first feed lines and second feed lines being one-to-one correspondingly arranged to form a plurality of feeding units being one-to-one corresponding to the radiators and feed current to the radiators; the coupling layer couples the current flowing through the feeding units to the radiation module; the coupling layer is provided with a plurality of first coupling slots and second coupling slots arranged in one-to-one correspondence, to form a plurality of coupling units.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims priority to Chinese Patent Application No. 202410229383.3 filed on Feb. 29, 2024, in China National Intellectual Property Administration, the contents of which are incorporated by reference herein.

FIELD

[0002] The subject matter herein generally relates to antenna technology field, and more particularly to an antenna array and a ground station.

BACKGROUND

[0003] Low-orbit satellite system (LEO) is a large satellite system composed of multiple satellites that can process real-time information. In the related technologies of low-orbit satellite systems, the antenna used to achieve communication between the ground station and the low-orbit satellite uses a circularly polarized antenna, which can avoid polarization loss of the signal on the transmission path and improve the reliability of the signal. The antenna can achieve circular polarization by setting two feed sources with a phase difference of 90 degrees between the two feed sources. Low-orbit satellite systems use antenna arrays to transmit and receive signals, and the performance of antenna arrays is easily affected by space, such as mutual influence between multiple antenna units, or mutual influence between two feed sources of the same antenna unit. Due to the limitation of antenna array area, antenna units or feed sources may be coupled and interfered with each other, which reduces the radiation efficiency of the antenna array and greatly limits the communication capability between satellites and ground stations or other satellites.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] Implementations of the present disclosure will now be described, by way of embodiments, with reference to the attached figures.

[0005] FIG. 1 is a top view diagram of an antenna array according to an embodiment of the present application.

[0006] FIG. 2 is a cross-sectional view diagram of a radiation unit of the antenna array obtained along a section line II-II of FIG. 1.

[0007] FIG. 3 is an exploded diagram of the radiation unit of FIG. 2.

[0008] FIG. 4 is a structural diagram of a feeding module of the radiation unit of FIG. 1.

[0009] FIG. 5 is a structural diagram of a coupling layer of a first specific embodiment of the present application.

[0010] FIG. 6 is a schematic diagram of a current path of the coupling layer shown in FIG. 5.

[0011] FIG. 7 is a scattering parameter (S-parameter) diagram of the radiation unit using the coupling layer of FIG. 5.

[0012] FIG. 8 is an antenna gain diagram of the radiation unit using the coupling layer of FIG. 5.

[0013] FIG. 9 is a structural diagram of a coupling layer of a second specific embodiment of the present application.

[0014] FIG. 10 is a schematic diagram of a current path of the coupling layer shown in FIG. 9.

[0015] FIG. 11 is a scattering parameter (S-parameter) diagram of the radiation unit using the coupling layer of FIG. 9.

[0016] FIG. **12** is an antenna gain diagram of the radiation unit using the coupling layer of FIG. **9**.
[0017] FIG. **13** is a structural diagram of a coupling layer of a third specific embodiment of the present application.
[0018] FIG. **14** is a scattering parameter (S-parameter) diagram of the radiation unit using the coupling layer of FIG. **13**.
[0019] FIG. **15** is an antenna gain diagram of the radiation unit using the coupling layer of FIG. **13**.
[0020] FIG. **16** is a structural diagram of a coupling layer of a fourth specific embodiment of the present application.
[0021] FIG. **17** is a scattering parameter (S-parameter) diagram of the radiation unit using the coupling layer of FIG. **16**.
[0022] FIG. **18** is an antenna gain diagram of the radiation unit using the coupling layer of FIG. **16**.
[0023] FIG. **19** is a structural diagram of a coupling layer of a fifth specific embodiment of the present application.
[0024] FIG. **20** is a scattering parameter (S-parameter) diagram of the radiation unit using the coupling layer of FIG. **19**.
[0025] FIG. **21** is an antenna gain diagram of the radiation unit using the coupling layer of FIG. **19**.
[0026] FIG. **22** is a structural diagram of a coupling layer of a sixth specific embodiment of the present application.
[0027] FIG. **23** is a scattering parameter (S-parameter) diagram of the radiation unit using the coupling layer of FIG. **22**.
[0028] FIG. **24** is an antenna gain diagram of the radiation unit using the coupling layer of FIG. **22**.
[0029] FIG. **25** is a structural diagram of a coupling layer of a seventh specific embodiment of the present application.
[0030] FIG. **26** is a scattering parameter (S-parameter) diagram of the radiation unit using the coupling layer of FIG. **25**.
[0031] FIG. **27** is an antenna gain diagram of the radiation unit using the coupling layer of FIG. **25**.
[0032] FIG. **28** is a structural diagram of a coupling layer of a eighth specific embodiment of the present application.
[0033] FIG. **29** is a scattering parameter (S-parameter) diagram of the radiation unit using the coupling layer of FIG. **28**.
[0034] FIG. **30** is an antenna gain diagram of the radiation unit using the coupling layer of FIG. **28**.
[0035] FIG. **31** is a structural diagram of a coupling layer of a ninth specific embodiment of the present application.
[0036] FIG. **32** is a schematic diagram of a current path of the coupling layer shown in FIG. **31**.
[0037] FIG. **33** is a scattering parameter (S-parameter) diagram of the radiation unit using the coupling layer of FIG. **31**.
[0038] FIG. **34** is a cross-sectional view diagram of the radiation unit according to another embodiment of the present application.
[0039] FIG. **35** is an exploded diagram of the radiation unit of FIG. **34**.
[0040] FIG. **36** is a block diagram of a radio frequency (RF) module according to an embodiment of the present application.
[0041] FIG. **37** is a block diagram of an electronic device according to an embodiment of the present application.

DETAILED DESCRIPTION

[0042] It will be appreciated that for simplicity and clarity of illustration, where appropriate, reference numerals have been repeated among the different figures to indicate corresponding or analogous elements. Additionally, numerous specific details are set forth in order to provide a thorough understanding of the embodiments described herein. However, it will be understood by those of ordinary skill in the art that the embodiments described herein can be practiced without these specific details. In other instances, methods, procedures and components have not been

described in detail so as not to obscure the related relevant feature being described. The drawings are not necessarily to scale and the proportions of certain parts may be exaggerated to better illustrate details and features. The description is not to be considered as limiting the scope of the embodiments described herein.

[0043] Several definitions that apply throughout this disclosure will now be presented.

[0044] The term “coupled” is defined as connected, whether directly or indirectly through intervening components, and is not necessarily limited to physical connections. The connection can be such that the objects are permanently connected or releasably connected. The term “substantially” is defined to be essentially conforming to the particular dimension, shape, or another word that “substantially” modifies, such that the component need not be exact. For example, “substantially cylindrical” means that the object resembles a cylinder, but can have one or more deviations from a true cylinder. The term “comprising” means “including, but not necessarily limited to”; it specifically indicates open-ended inclusion or membership in a so-described combination, group, series, and the like.

[0045] Low-orbit satellite system (LEO) is a large satellite system composed of multiple satellites that can process real-time information. In the related technologies of low-orbit satellite systems, the antenna gain used to realize communication between ground stations and low-orbit satellites is low and cannot well meet user needs, which greatly limits the communication ability of satellites to communicate with ground stations or other satellites.

[0046] Based on this, this application provides an antenna array that can be applied to ground stations to achieve communication among low-orbit satellites, and the antenna array has high antenna gain.

Embodiment I

[0047] Referring to FIG. 1, FIG. 1 illustrates an overall schematic diagram of an antenna array **10** provided by the present application. The antenna array **10** includes at least one radiation module and a feeding module arranged in a stack. The radiation module is used to transmit or receive signals, the feeding module is used to provide feed power to the radiation module. It can be understood that the at least one radiation module and the feeding module form the antenna array **10** including a plurality of radiation units **11**. In the following embodiments, a structural schematic diagram of a radiation unit **11** in the antenna array **10** is used as an example to illustrate the specific structure of the antenna array **10**.

[0048] Referring to FIG. 2, FIG. 2 illustrates a cross-sectional view diagram of one of the plurality of radiation units **11** of the antenna array **10**. In the antenna array **10** of the present embodiment, the at least one radiation module includes a first radiation module **110**. Referring to FIG. 3, the first radiation module **110** includes a first radiation layer **111** and a first dielectric layer **112** arranged in stack. The first radiation layer **111** includes a plurality of first radiators **1111**, each of the plurality of first radiators **1111** is configured to radiate signals. In some embodiment, the first radiation layer **111** may be a printed circuit board, or a board made of ceramic material or plastic material board, or the like. The first radiator **1111** may be a metal coating formed on the first radiation layer **111** or other sheets made of conductive materials. The first radiator **1111** is formed on a surface of the first radiation layer **111** away from the first dielectric layer **112**. The first dielectric layer **112** is made of non-conductive material. For instance, in some embodiments, the first dielectric layer **112** may be made of a material with a dielectric material coefficient of about 2.4. Specifically, the first dielectric layer **112** may be made of a ceramic material or a plastic material. Specifically, in some embodiments, the first radiator **1111** is a substantially circular copper sheet. The first radiation layer **111** and the first dielectric layer **112** are both substantially square. An area of each of the first radiating layer **111** and the first dielectric layer **112** is greater than an area of the first radiator **1111**. It is understandable that the first radiator **111** may be made of conductive metal materials such as copper, aluminum, and silver.

[0049] Referring to FIGS. 3 and 4, the antenna array **10** further includes a feeding module **130**. The

feeding module **130** is arranged on a side of the first radiation module **110** closes to the first dielectric layer **112**. The feeding module **130** includes a feeding layer **133** and a coupling layer **131** arranged in a stack. The feeding layer **133** is provided with a plurality of first feed lines **1331** and a plurality of second feed lines **1332**. The first feed lines **1331** and the second feed lines **1332** are arranged in one-to-one correspondence, forming a plurality of feeding units. The feeding units correspond to the first radiators **111** on the first radiation layers **111** one-to-one, the feeding units are used to feed currents for the corresponding first radiators **111**. In an embodiment of the present application, the one-to-one correspondence arrangement means that in a projection along the Z-axis direction of the antenna array, the projected areas of the two structures arranged in one-to-one correspondence at least partially overlap. Thus, the feeding units and the first radiators **111** on the first radiation layers **111** are arranged in one-to-one correspondence. That is to say, in the projection along the Z-axis direction of the antenna array **10**, a projected area of the feeding units and a projected area of the first radiators **111** at least partially overlap each other. For instance, in the projection along the Z-axis direction of the antenna array **10**, the projected area of the feeding units completely overlaps the projected area of the first radiators **111**, or the projected area of the feeding units and the projected area of the first radiators **111** partially overlap each other. Based on this, the plurality of first feed lines **1331** and the plurality of second feed lines **1332** are in one-to-one correspondence arrangement means that in the feeding layer **133**, the first feed lines **1331** and the second feed lines **1332** corresponding to a same first radiator **111**, that is, in the projection along the Z-axis direction of the antenna array **10**, a projected area of the first radiator **111** and a projected area of the first feed line **1331** and the second feed line **1332** overlap or at least partially overlap each other. In an embodiment of the present application, there is an included angle between the first feed line **1331** and the second feed line **1332**, so as to generate polarized waves in two different directions, especially horizontally polarized waves and vertically polarized waves.

[0050] The coupling layer **131** is arranged on a side of the feeding layer **133** closes to the first radiation module **110**, that is, the coupling layer **131** is arranged close to the first dielectric layer **112** of the first radiation module **110**, specifically arranged on a side of the first dielectric layer **112** away from the first radiation layer **111**. The coupling layer **131** is used to couple a current signal flowing through the feed line unit to the first radiation module **110**.

[0051] The coupling layer **131** is provided with a plurality of first coupling slots **1311** and a plurality of second coupling slots **1312** (see FIG. 4). The plurality of first coupling slots **1311** and the plurality of second coupling slots **1312** are arranged in one-to-one correspondence to form a plurality of coupling units. The first coupling slot **1311** and the second coupling slot **1312** of the coupling unit are spaced apart from each other. Thus, when the coupling layer **131** couples the current signal flowing through the feeding units to the first radiation module **110**, the coupling layer **131** changes the path and direction of the current of the coupling layer **131** through the first coupling slots **1311** and the second coupling slots **1312**, which can reduce a coupling interference between the first feeding line **1331** and the second feeding line **1332**, improve an isolation between the first feeding line **1331** and the second feeding line **1332**, and reduce an interference influence of the antenna array **10**.

[0052] In some embodiments, the plurality of first coupling slots **1311** and the plurality of second coupling slots **1312** have different directions, so that an extension line of the first coupling slots **1311** and an extension line of the second coupling slots **1312** of the coupling unit intersect.

[0053] In some embodiments, the first coupling slot **1311** and the second coupling slot **1312** of the coupling unit are perpendicular to each other and spaced apart (as shown in FIGS. 3 and 4). It can be understood that when the first coupling slot **1311** and the second coupling slot **1312** are perpendicular to each other, an angle between the extension line of the first coupling slot **1311** and the extension line of the second coupling slot **1312** is 90°. In some other embodiments, the angle between the extension line of the first coupling slot **1311** and the extension line of the second coupling slot **1312** can be 30°, 45°, 55°, etc., a specific value of the angle between the extension

line of the first coupling slot **1311** and the extension line of the second coupling slot **1312** is not limited by the embodiments of the present application.

[0054] Further, at least one end of the first coupling slot **1311** and the second coupling slot **1312** of the coupling unit is formed with a port shape. It can be understood that at least one end may be one end, two ends, three ends, or four ends of the first coupling slot **1311** and the second coupling slot **1312** of the coupling unit. The port shape may be formed with an angle. When the end of the first coupling slot **1311** is formed with a port shape, the angle formed by the port shape of the first coupling slot **1311** may be toward or away from the first coupling slot **1311**. When the end of the second coupling slot **1312** is formed with a port shape, the angle formed by the port shape of the second coupling slot **1312** may be toward or away from the second coupling slot **1312**.

[0055] In some embodiments, the plurality of first coupling slots **1311** and the plurality of second coupling slots **1312** have different shapes. For example, the first coupling slot **1311** is in the shape of a unidirectional arrow as a whole, and the second coupling slot **1312** is in the shape of a bidirectional arrow as a whole. Or, the first coupling slot **1311** is in the shape of a bidirectional arrow as a whole, and the second coupling slot **1312** is in the shape of a unidirectional arrow as a whole. Or, the first coupling slot **1311** is in the shape of an H-shaped as a whole, and the second coupling slot **1312** is in the shape of a unidirectional arrow or a bidirectional arrow as a whole, and vice versa. Another example, the first coupling slot **1311** is in the shape of a unidirectional arrow as a whole, and the second coupling slot **1312** is in the shape of a long strip shaped as a whole. The embodiments of the present application do not limit these.

[0056] Of course, in other embodiments, the plurality of first coupling slots **1311** and the plurality of second coupling slots **1312** have a same shape. For example, the first coupling slot **1311** and the second coupling slot **1312** are both unidirectional arrows or bidirectional arrows.

[0057] Furthermore, each coupling unit is disposed in one-to-one correspondence with the feeding unit, the first coupling slot **1311** is disposed in correspondence with the corresponding first feeding line **1331**, and the second coupling slot **1312** is disposed in correspondence with the corresponding second feeding line **1332**. In the embodiment, the shape of the first coupling slot **1311** and the projection shape of the first feeding line **1331** in the Z-axis direction of the antenna array **10** may be the same or different, and the shape of the second coupling slot **1312** and the projection shape of the second feeding line **1332** in the Z-axis direction of the antenna array **10** may be the same or different, but the first coupling slot **1311** and the second coupling slot **1312** need to be narrow and long as a whole to achieve the coupling effect, such as an ellipse. Thus, the coupling layer **131** couples the current signal flowing through the feeding unit to the first radiation module **110** through the first coupling slot **1311** and the second coupling slot **1312**. In other embodiments, the coupling layer **131** may be replaced by a first coupling layer and a second coupling layer (not shown), and the first coupling layer is provided with a first coupling slot corresponding to the first feeding line **1331**, and the second coupling layer is provided with a second coupling slot corresponding to the second feeding line **1332**. Thus, the current signal flowing through the feeding unit can also be coupled to the first radiation module **110** through the design of the first coupling layer and the second coupling layer.

[0058] In some embodiments, along the projection direction of the Z axis of the antenna array **10**, the projection of the first coupling slot **1311** and the projection of the corresponding first feeding line **1331** are perpendicular to each other, and the projection of the second coupling slot **1312** and the projection of the corresponding second feeding line **1332** are perpendicular to each other, so the current signals of the first feeding line **1331** and the second feeding line **1332** can be better coupled to the first radiation module **110** through the first coupling slot **1311** and the second coupling slot **1312**.

[0059] It can be understood that if a distance between a midpoint of the first coupling slot **1311** and the second coupling slot **1312** is too close, the first coupling slot **1311** and the second coupling slot **1312** will interfere with each other, while if the distance is too far, the size of the antenna array **10**

will be too large. Based on this, in the embodiment of the present application, the distance is set to one-sixth of a wavelength, and the distance between an end of the first coupling slot **1311** close to the corresponding second coupling slot **1312** and the second coupling slot **1312** is set to one-twelfth of the wavelength. In some specific examples, one-sixth of the wavelength is about 4.6 millimeters (mm), and one-twelfth of the wavelength is about 1.65 mm. In some other specific examples, one-sixth of the wavelength is 4 mm, and one-twelfth of the wavelength is 2 mm.

[0060] Furthermore, in some embodiments, the feeding module **130** further includes a first cavity layer **132**, a second cavity layer **134**, and a ground layer **135**. The first cavity layer **132** is disposed between the feeding layer **133** and the coupling layer **131**, and the second cavity layer **134** is disposed between the feeding layer **133** and the ground layer **135**. The first cavity layer **132** is provided with a plurality of first through cavities **1321**, and the second cavity layer **134** is provided with a plurality of second through cavities **1341**. The first through cavities **1321** and the second through cavities **1341** are both provided in one-to-one correspondence with the first radiators **1111**. A projection area of each of the first through cavities **1321** and each of the second through cavities **1341** in the projection direction along the Z axis of the antenna array **10** at least partially overlap with the projection area of the corresponding first radiator **1111** in the projection direction along the Z axis.

[0061] In the embodiment of the present application, the first cavity layer **132** is disposed between the feeding layer **133** and the coupling layer **131**, and the second cavity layer **134** is disposed between the feeding layer **133** and the ground layer **135**, so as to isolate the coupling layer **131**, the feeding layer **133**, and the ground layer **135**, and also to increase the antenna height of the antenna array **10**, thereby increasing the antenna gain of the antenna array **10**.

[0062] The feeding layer **133** is provided with a plurality of first receiving grooves **1333** (see FIG. 4), and the first receiving grooves **1333** penetrate the feeding layer **133**. Each first feed line **1331** and each second feed line **1332** are respectively disposed in the corresponding first receiving groove **1333**.

[0063] Further, in some embodiments, the feeding module **130** further includes a plurality of phase couplers **1334**. The phase couplers **1334** are used to realize signal transmission between the antenna array **10** and signal transceivers (see FIG. 36, including a transmitter **150** and a receiver **160**). The phase couplers **1334** may be arranged on the feeding layer **133**. In this embodiment, the phase couplers **1334** are arranged in the first receiving groove **1333** of the feeding layer **133**. Specifically, the first feed line **1331** and the second feed line **1332** are both generally elongated microstrip lines. The phase coupler **1333** is substantially a square ring-shaped metal ring. The first receiving groove **1333** includes a first groove **13331** and a second groove **13332** that communicate with each other. The phase coupler **1334** is arranged in the first groove **13331**, the first feed line **1331** and the second feed line **1332** are arranged in the second groove **13332**. The shapes of the first groove **13331** and the second groove **13332** can be set according to the relationship between the first feed line **1331** and the second feed line **1332**. Specifically, when the first feed line **1331** and the second feed line **1332** need to be set to be equal in length and extend in different directions, the shape of the first groove **13331** can be set to be square, and the shape of the second groove **13332** can be set to be circular or elliptical. Of course, it is understood that the specific shape of the first receiving groove **1333** in the embodiment of the present application is not limited thereto and can be set according to actual needs.

[0064] The phase coupler **1334** includes a signal receiving end RX, a signal transmitting end TX, a first signal feeding end F1, and a second signal feeding end F2. The signal receiving end RX and the signal transmitting end TX are arranged on one side of the phase coupler **1333**, the first signal feeding end F1 and the second signal feeding end F2 are arranged on the other side of the phase coupler **1333** away from the signal receiving end RX and the signal transmitting end TX. Each phase coupler **1334** is used to connect correspondingly to the first feed line **1331** and the second feed line **1332**, to output a first feed signal to the first feed line **1331**, and to output a second feed

signal to the second feed line **1332**. Phases of the first feed signal and the second feed signal may be the same or different. For instance, in this embodiment, the phases of the first feed signal and the second feed signal may differ by 90° or -90° . In this way, circularly polarized waves can be excited in the radiation unit **11**. The signal receiving end RX is used to connect to the receiver **160** in the signal transceiver, the signal transmitting end TX is used to connect to the transmitter **150** in the signal transceiver.

[0065] An extending direction of the first feed line **1331** in the feeding layer **133** does not completely overlap with an extending direction of the corresponding second feed line **1332** in the feeding layer **133**. The first feed line **1331** is used to generate a first polarized wave, and the second feed line **1332** is used to generate a second polarized wave. For example, in this embodiment, on the feeding layer **133**, the first feed line **1331** extends in a first direction, and the second feed line **1332** extends in a second direction, the first direction is a direction away from the phase coupler **1334**, the second direction is a direction away from the first feed line **1331**, and the first direction and the second direction are perpendicular. In addition, when the phase difference between the first feed signal and the second feed signal is 90° , the first feed line **1331** and the second feed line **1332** can cooperatively excite a dual circular polarization signal, which is beneficial for communication with the low-orbit satellite. Furthermore, when a radiation pattern of the signal transmitted by the signal receiving end RX and a radiation pattern of the signal transmitted by the signal transmitting end TX are circular polarization patterns opposite to each other, the antenna array **10** can also realize simultaneous signal reception and transmission based on dual circular polarization waves through different radiation units **11**.

[0066] More specifically, the first feed line **1331** can include a first feed section (not labeled in the figures), a second feed section (not labeled in the figures), and a third feed section (not labeled in the figures). A first end of the first feed section is connected to the first signal feeding end F1, a second end of the first feed section extends along the first direction and is connected to a first end of the second feed section, a second end of the second feed section extends along a direction opposite to the second direction and is connected to a first end of the third feed section, a second end of the third feed section extends along the first direction. The second feed line **1332** can include a fourth feed section (not labeled in the figures) and a fifth feed section (not labeled in the figures). A first end of the fourth feed section is connected to the second signal feeding end F2, a second end of the fourth feed section extends along the first direction and is connected to a first end of the fifth feed section, a second end of the fifth feed section extends along the second direction. Along the projection direction of the Z-axis of the antenna array **10**, the projection of the first coupling slot **1311** and the projection of the corresponding third feed section of the first feed line **1331** are perpendicular to each other, and the projection of the second coupling slot **1312** and the projection of the corresponding fifth feed section of the second feed line **1332** are perpendicular to each other.

[0067] It can be understood that a length of the first feed line **1331** is a same with a length of the second feed line **1332**.

[0068] The ground layer **135** is arranged on a side of the second cavity layer **134** away from the feeding layer **133**. Understandably, the ground layer **135** may be a metal coating arranged on a printed circuit board. The metal coating can be arranged on a side of the ground layer **135** away from the second cavity layer **134**. In the embodiment, the coupling layer **131**, the first cavity layer **132**, the feeding layer **133**, the second cavity layer **134**, and the ground layer **135** are each provided with through holes **1312** (not labeled in the figures), the through holes of each layer on the feeding module **130** are connected in sequence, and finally connected to the metal coating on the ground layer **135** to be grounded.

[0069] Referring to FIG. 4, in the embodiment, the first feed line **1331** is connected to the first signal feeding end F1 of the phase coupler **1334** through a metal connecting piece. The second feed line **1332** is connected to the second signal feeding end F2 of the phase coupler **1334** through a

metal connecting piece. The feeding layer **133**, the second cavity layer **134**, and the ground layer **135** may be provided with a first connection hole and a second connection hole (not shown in the figures), so that the signal receiving end RX and the signal transmitting end TX of the phase coupler **1334** pass through the feeding layer **133**, the second cavity layer **134**, and the ground layer **135** through the first connection hole and the second connection hole respectively, so as to be connected to the transmitter **150** and the receiver **160** (see FIG. **36**) in the signal transceiver. In other embodiments, the through hole, the first connection hole, and the second connection hole may also be replaced by a feeding probe, the present application does not limit the manner in which the antenna array **10** realizes electrical connection in the multi-layer structure.

[0070] Referring to FIG. **3** again, in some embodiments, the feeding module **130** further includes a first dielectric body **136**, a second dielectric body **137**, a third dielectric body **138**, and a fourth dielectric body **139**. The first dielectric body **136** is arranged between the coupling layer **131** and the first cavity layer **132**, the second dielectric body **137** is arranged between the first cavity layer **132** and the feeding layer **133**, the third dielectric body **138** is arranged between the feeding layer **133** and the second cavity layer **134**, the fourth dielectric body **139** is arranged between the second cavity layer **134** and the ground layer **135**. The first dielectric body **136**, the second dielectric body **137**, the third dielectric body **138**, and the fourth dielectric body **139** may also be provided with the through hole, the first connection hole, and the second connection. It can be understood that the first dielectric body **136**, the second dielectric body **137**, the third dielectric body **138**, and the fourth dielectric body **139** can be made of a material with a dielectric constant of about 2.4.

[0071] It should be known that the antenna array **10** is also connected to the phase modulation module (not shown in the figures). The phase modulation module is used to adjust phases of the transmitting signal and the receiving signal of the antenna array **10** to achieve high-efficiency communication between the device where the antenna array **10** is applied in and the low-orbit satellite through beam forming technology. For instance, the phase modulation module may include a control unit, a combiner, an attenuator, a power amplifier, a low-noise amplifier, etc. this application is not limited to a specific circuit structure of the phase modulation module.

[0072] In the embodiment, a working principle of the antenna array **10** is roughly as follows:

[0073] When the radiation unit **11** shown in FIG. **2** is used to transmit signals, the transmitter **150** in the signal transceiver feeds radio frequency signals to the first feed line **1331** and the second feed line **1332** through the signal transmitting end TX. Furthermore, since the extension directions of the first feed line **1331** and the second feed line **1332** are perpendicular to each other, a current path flowing through the first feed line **1331** and a current path flowing through the second feed line **1332** are orthogonal to each other. After the first feed line **1331** and the second feed line **1332** receive the feed signal through the first signal feed end F1 and the second signal feed end F2 respectively, in the Z-axis projection direction of the antenna array **10**, the first feed line **1331** and the second feed line **1332** stagger each other at an angle. In this way, the first feed line **1331** and the second feed line **1332** can generate a first polarized wave and a second polarized wave respectively, especially generating horizontally polarized waves and vertically polarized waves. Additionally, through the phase coupler **1334**, the first polarized wave generated by the first feed line **1331** and the second polarized wave generated by the second feed line **1332** are simultaneously coupled and fed to the first radiator **1111** through the first coupling slots **1311** and the second coupling slots **1312** on the coupling layer **131**, causing the first radiator **1111** to transmit a left-hand circularly polarized wave or a right-hand circularly polarized wave outward. In some embodiments, an operating frequency band when the antenna array **10** is used to transmit signals may be 14 GHz-14.5 GHz.

[0074] When the radiation unit **11** shown in FIG. **2** is used to receive signals, the first radiator **1111** receives electromagnetic waves from the outside and converts them into electrical signals, the first radiator **1111** couples the electrical signals to the first feed line **1331** and the second feed line **1332** through the first coupling slot **1311** and the second coupling slot **1312**, the first feed line **1331** and

the second feed line **1332** further feed back the electrical signals to the receiver **160** in the signal transceiver through the phase coupler **1334**. Since the first coupling slot **1311** and the second coupling slot **1312** are perpendicular to and spaced apart from each other, the extension directions of the first feed line **1331** and the second feed line **1332** are perpendicular to each other, so the first feed line **1331** can couple to generate the first polarized waves, and the second feed line **1332** can couple to generate the second polarized waves, the polarization directions of the first polarized waves and the second polarized waves are perpendicular to each other. Therefore, the current path of the current signal generated by the first feed line **1331** according to the first polarized wave and the current path of the current signal generated by the second feed line **1332** according to the second polarized wave are orthogonal to each other. In this way, the first feed line **1331** and the second feed line **1332** transmit the generated current signal to the signal receiving end RX through the first signal feeding end F1 and the second signal feeding end F2 respectively, and then transmit the received current signal to the receiver **160** in the signal transceiver through the signal receiving end RX to realize the reception of the antenna signal. In some embodiments, an operating frequency band when the antenna array **10** is used to receive signals may be 10.7 GHz-12.5 GHz. [0075] In some embodiments, the radiation field type of the signal transmitted by the signal receiving end RX of the phase coupler **1334** and the radiation field type of the signal transmitted by the signal transmitting end TX are mutually opposite circular polarization field types. That is to say, the radiation unit **11** may transmit one of the left-hand circularly polarized wave or the right-hand circularly polarized wave when transmitting a signal, and the radiating unit **11** may transmit the other of the left-hand circularly polarized wave or the right-hand circularly polarized wave when receiving a signal. In some other embodiments, a radio frequency switch can also be set to control the radiation unit **11** to transmit the left-hand circular polarized wave or the right-hand circular polarized wave when receiving a signal, and to control the radiation unit **11** to transmit the left-hand circular polarized wave or the right-hand circularly polarized wave when transmitting a signal.

[0076] It can be understood that since the antenna array **10** includes a plurality of radiating units **11** (for example, 1024 radiating units **11**), in some embodiments, the phase modulation module can control some of the radiating units **11** in the antenna array **10** to transmit signals, at the same time, other radiating units **11** in the antenna array **10** are controlled to receive signals. In this way, the antenna array **10** can simultaneously receive and transmit signals from low-orbit satellites, thereby improving communication efficiency with low-orbit satellites.

[0077] It can be understood that in each radiating unit **11** of the antenna array **10**, the projected area of the first radiator **1111** in the Z-axis direction of the antenna array **10** completely covers the projected area of the first coupling slot **1311** and the second coupling slot **1312** in the Z-axis direction of the antenna array **10**. So that the energy of the first feed line **1331** and the second feed line **1332** can be coupled to the first radiator **1111** as much as possible.

[0078] Referring to FIG. 3 again, in some embodiments, the antenna array **10** further includes a protective layer **120**. The protective layer **120** is disposed on the side of the first radiation layer **111** away from the first dielectric layer **112** to protect the antenna array **10** from an influence of sunlight, rain, and dust, so as to improve the working stability of the antenna array **10**. In this embodiment, the protective layer **120** is also provided with a protective cavity **121** corresponding to the first radiator **1111**. For example, the protective cavity **121** may be formed by an inward recess of a side of the protective layer **120** closes to the first radiator **1111**, and the protective cavity **121** may be substantially cylindrical. In this way, the weight of the antenna array **10** can be reduced.

[0079] Further, in the antenna array **10**, the first radiating layer **111**, the first dielectric layer **112**, the first cavity layer **132**, the second cavity layer **134**, the first dielectric body **136**, the second dielectric body **137**, the third dielectric body **138**, and the fourth dielectric body **139** can also define through holes at positions that do not correspond to the first radiator **1111**. In this way, the weight

of the antenna array **10** can be further reduced.

[0080] It can be understood that in the antenna array **10**, each adjacent two-layer structure can be connected by adhesive, this application does not limit the specific type of adhesive.

[0081] As shown in FIG. **3** and FIG. **4**, in the first specific example of the embodiment of the present application, one end of the first coupling slot **1311** and the second coupling slot **1312** is at least formed with a port shape. An angle formed by the port shape at one end of the first coupling slot **1311** is oriented toward the first coupling slot **1311**, and the first coupling slot **1311** is in the shape of a unidirectional arrow as a whole.

[0082] Specifically, as shown in FIG. **5**, one end of the first coupling slot **1311a** is formed with a port shape, the end of the first coupling slot **1311a** formed with the port shape is close to the second coupling slot **1312**, the port shape is formed with an angle, and the angle is toward the first coupling slot **1311a**. No port shape with an angle is formed at both ends of the second coupling slot **1312**. That is, in this example, the port shape is only formed at one end of the first coupling slot **1311a**.

[0083] Specifically, the first coupling slot **1311a** includes a first main slot **13110a** which is narrow and long, a first sub-slot **13111a** and a second sub-slot **13112a** which are connected to the first main slot **13110a** are formed at both ends of the first main slot **13110a**, respectively. The first sub-slot **13111a** is close to the second coupling slot **1312** and an angle is formed as a whole, and the angle is toward the first main slot **13110a**. The second sub-slot **13112a** is vertically connected to the first main slot **13110a**, so that the first coupling slot **1311a** is in the shape of a unidirectional arrow as a whole. The second coupling slot **1312a** includes a second main slot **13120a** which is narrow and long, a third sub-slot **13121a** and a fourth sub-slot **13122a** which are connected to the second main slot **13120a** are respectively formed at both ends of the second main slot **13120a**, the third sub-slot **13121a** and the fourth sub-slot **13122a** are vertically connected to the second main slot **13120a**, so that the second coupling slot **1312a** is H-shaped as a whole. An extension line of the first main slot **13110a** of the first coupling slot **1311a** and the second main slot **13120a** of the second coupling slot **1312a** are perpendicular to each other.

[0084] Referring to FIG. **6**, FIG. **6** is a schematic diagram of current paths of the coupling layer **131a** in the example of FIG. **5**. According to FIG. **6**, the current path and direction of the current on the coupling layer **131a** are changed by the unidirectional arrow shape of the first coupling slot **1311a**, the current is concentrated near the port shape formed by the first coupling slot **1311a**, thereby reducing the coupling interference between the first feed line **1331** and the second feed line **1332**, improving the isolation between the first feed line **1331** and the second feed line **1332**, and thus reducing the interference effect of the antenna array **10**.

[0085] Referring to FIG. **7**, FIG. **7** is a scattering parameter (S-parameter) curve diagram of the radiation unit **11** using FIG. **5** as a coupling layer. Curve **S71** represents a **S11** value of the signal receiving end RX when the radiation unit **11** is used to receive signals; curve **S72** represents a **S11** value of the signal transmitting end TX when the radiation unit **11** is used to transmit signals; curve **S73** represents a **S12** value between the signal receiving end RX and the signal transmitting end TX. As can be seen from FIG. **7**, the radiation unit **11** has a good reflection coefficient when transmitting or receiving signals, and the **S12** value between the signal receiving end RX and the signal transmitting end TX of the radiation unit **11** can reach below -50 dB, which can effectively reduce the coupling effect between the first feed line **1331** and the second feed line **1332**, so that the first feed line **1331** and the second feed line **1332** have better isolation.

[0086] Referring to FIG. **8**, FIG. **8** is an antenna gain curve diagram of the radiation unit **11** using FIG. **5** as a coupling layer. A gain value of the signal transmitting end TX when the radiation unit **11** is used to transmit signals is substantially coincident with a gain value of the signal receiving end RX when the radiation unit **11** is used to receive signals. As can be seen from FIG. **8**, the radiation unit **11** of this example has a high antenna gain, which meets the antenna working requirements. The gain value of the signal transmitting end TX when the radiation unit **11** is used to

transmit signals is substantially the same as the gain value of the signal receiving end RX when the radiation unit **11** is used to receive signals.

[0087] Referring to FIG. 9, FIG. 9 shows a schematic diagram of the structure of the coupling layer **131b** of a second specific example of the embodiment. As shown in FIG. 9, both ends of the first coupling slot **1311** and the second coupling slot **1312** are formed with the port shapes, and the port shapes form an angle, that is, the port shapes are formed at all ends of the first coupling slot **1311** and the second coupling slot **1312**. The angle formed by the port shapes at both ends of the first coupling slot **1311** is oriented toward the first coupling slot **1311**, and the angle formed by the port shapes at both ends of the second coupling slot **1312** is oriented toward the second coupling slot **1312**. Thus, the first coupling slot **1311** is in the shape of a bidirectional arrow as a whole, and the second coupling slot **1312** is also in the shape of a bidirectional arrow as a whole.

[0088] Specifically, the first coupling slot **1311b** includes a first main slot **13110b** which is narrow and long, a first sub-slot **13111b** and a second sub-slot **13112b** which are connected to the first main slot **13110b** are respectively formed at both ends of the first main slot **13110b**, the first sub-slot **13111b** and the second sub-slot **13112b** are formed with an angle as a whole, and the angle is toward the first main slot **13110b**, the first sub-slot **13111b** and the second sub-slot **13112b** are symmetrically arranged along a center of the first main slot **13110b**. The second coupling slot **1312b** includes a second main slot **13120b** which is narrow and long, a third sub-slot **13121b** and a fourth sub-slot **13122b** that are connected to the second main slot **13120b** are respectively formed at both ends of the second main slot **13120b**, the third sub-slot **13121b** and the fourth sub-slot **13122b** are formed with an angle as a whole, and the angle is toward the second main slot **13120b**, the third sub-slot **13121b** and the fourth sub-slot **13122b** are symmetrically arranged along a center of the second main slot **13120b**. Therefore, the first coupling slot **1311b** and the second coupling slot **1312b** are in the shape of a bidirectional arrow as a whole.

[0089] An extension line of the first main slot **13110b** of the first coupling slot **1311b** and the second main slot **13120b** of the second coupling slot **1312b** are perpendicular to each other. An intersection point of the extension line of the first main slot **13110b** of the first coupling slot **1311b** and the second main slot **13120b** of the second coupling slot **1312b** may be a midpoint of the second main slot **13120b**. Of course, it is understood that, in some other embodiments, the first sub-slot **13111b** and the second sub-slot **13112b** may not be symmetrically arranged along the center of the first main slot **13110b**, and the third sub-slot **13121b** and the fourth sub-slot **13122b** may not be symmetrically arranged along the center of the second main slot **13120b**.

[0090] Referring to FIG. 10, FIG. 10 is a schematic diagram of current paths of the coupling layer **131b** in the example of FIG. 9. According to FIG. 10, the current path and direction of the current on the coupling layer **131b** are changed by the bidirectional arrow shape of the first coupling slot **1311b** and the second coupling slot **1312b**, the current is concentrated near the port shape formed by the first coupling slot **1311b** and the second coupling slot **1312b**, thereby reducing the coupling interference between the first feed line **1331** and the second feed line **1332**, improving the isolation between the first feed line **1331** and the second feed line **1332**, and thus reducing the interference effect of the antenna array **10**.

[0091] Referring to FIG. 11, FIG. 11 is a scattering parameter (S-parameter) curve diagram of the radiation unit **11** using FIG. 9 as a coupling layer. Curve **S11** represents a **S11** value of the signal receiving end RX when the radiation unit **11** is used to receive signals; curve **S112** represents a **S11** value of the signal transmitting end TX when the radiation unit **11** is used to transmit signals; curve **S113** represents a **S12** value between the signal receiving end RX and the signal transmitting end TX. As can be seen from FIG. 11, the radiation unit **11** has a good reflection coefficient when transmitting or receiving signals, and the **S12** value between the signal receiving end RX and the signal transmitting end TX of the radiation unit **11** can reach below -50 dB, which can effectively reduce the coupling effect between the first feed line **1331** and the second feed line **1332**, so that the first feed line **1331** and the second feed line **1332** have better isolation.

[0092] Referring to FIG. 12, FIG. 12 is an antenna gain curve diagram of the radiation unit 11 using FIG. 9 as a coupling layer. A gain value of the signal transmitting end TX when the radiation unit 11 is used to transmit signals is substantially coincident with a gain value of the signal receiving end RX when the radiation unit 11 is used to receive signals. As can be seen from FIG. 12, the radiation unit 11 of this example has a high antenna gain, which meets the antenna working requirements.

[0093] Referring to FIG. 13, FIG. 13 shows a schematic diagram of the structure of the coupling layer 131c of a second specific example of the embodiment. As shown in FIG. 13, both ends of the first coupling slot 1311c are formed with the port shapes, and the port shapes form an angle, the angle is oriented toward the first coupling slot 1311c. Both ends of the second coupling slot 1312c do not form the port shapes with an angle.

[0094] Specifically, the first coupling slot 1311c includes a first main slot 13110c which is narrow and long, a first sub-slot 13111c and a second sub-slot 13112c which are connected to the first main slot 13110c are respectively formed at both ends of the first main slot 13110c, the first sub-slot 13111c and the second sub-slot 13112c are formed with an angle as a whole, and the angle is toward the first main slot 13110c, the first sub-slot 13111c and the second sub-slot 13112c are symmetrically arranged along a center of the first main slot 13110c, so the first coupling slot 1311c is in the shape of the bidirectional arrow as a whole. The second coupling slot 1312c includes a second main slot 13120c which is narrow and long, a third sub-slot 13121c and a fourth sub-slot 13122c that are connected to the second main slot 13120c are respectively formed at both ends of the second main slot 13120c, the third sub-slot 13121c and the fourth sub-slot 13122c are both vertically connected to the second main slot 13120c, so that the second coupling slot 1312c is H-shaped as a whole. An extension line of the first main slot 13110c of the first coupling slot 1311c and the second main slot 13120c of the second coupling slot 1312c are perpendicular to each other.

[0095] Referring to FIG. 14, FIG. 14 is a scattering parameter (S-parameter) curve diagram of the radiation unit 11 using FIG. 13 as a coupling layer. Curve S141 represents a S11 value of the signal receiving end RX when the radiation unit 11 is used to receive signals; curve S142 represents a S11 value of the signal transmitting end TX when the radiation unit 11 is used to transmit signals; curve S143 represents a S12 value between the signal receiving end RX and the signal transmitting end TX. As can be seen from FIG. 14, the radiation unit 11 has a good reflection coefficient when transmitting or receiving signals, and the S12 value between the signal receiving end RX and the signal transmitting end TX of the radiation unit 11 can reach below -50 dB, which can effectively reduce the coupling effect between the first feed line 1331 and the second feed line 1332, so that the first feed line 1331 and the second feed line 1332 have better isolation.

[0096] Referring to FIG. 15, FIG. 15 is an antenna gain curve diagram of the radiation unit 11 using FIG. 13 as a coupling layer. A gain value of the signal transmitting end TX when the radiation unit 11 is used to transmit signals is substantially coincident with a gain value of the signal receiving end RX when the radiation unit 11 is used to receive signals. As can be seen from FIG. 15, the radiation unit 11 of this example has a high antenna gain, which meets the antenna working requirements.

[0097] Referring to FIG. 16, FIG. 16 shows a schematic diagram of the structure of the coupling layer 131d of a fourth specific example of the embodiment. As shown in FIG. 16, both ends of the first coupling slot 1311d do not form the port shapes with an angle. Both ends of the second coupling slot 1312d are formed with the port shapes, and the port shapes form an angle, the angle is oriented toward the second coupling slot 1312d.

[0098] Specifically, the first coupling slot 1311d includes a first main slot 13110d which is narrow and long, a first sub-slot 13111d and a second sub-slot 13112d which are connected to the first main slot 13110d are respectively formed at both ends of the first main slot 13110d, the first sub-slot 13111d and the second sub-slot 13112d are both vertically connected to the first main slot 13110d, so that the first coupling slot 1311d is H-shaped as a whole. The second coupling slot

1312d includes a second main slot **13120d** which is narrow and long, a third sub-slot **13121d** and a fourth sub-slot **13122d** that are connected to the second main slot **13120d** are respectively formed at both ends of the second main slot **13120d**, the third sub-slot **13121d** and the fourth sub-slot **13122d** are formed with an angle as a whole, and the angle is toward the second main slot **13120d**, the third sub-slot **13121d** and the fourth sub-slot **13122d** are symmetrically arranged along a center of the second main slot **13120d** so the second coupling slot **1312d** is in the shape of the bidirectional arrow as a whole. An extension line of the first main slot **13110d** of the first coupling slot **1311d** and the second main slot **13120d** of the second coupling slot **1312d** are perpendicular to each other.

[0099] Referring to FIG. 17, FIG. 17 is a scattering parameter (S-parameter) curve diagram of the radiation unit **11** using FIG. 16 as a coupling layer. Curve **S171** represents a **S11** value of the signal receiving end RX when the radiation unit **11** is used to receive signals; curve **S172** represents a **S11** value of the signal transmitting end TX when the radiation unit **11** is used to transmit signals; curve **S173** represents a **S12** value between the signal receiving end RX and the signal transmitting end TX. As can be seen from FIG. 17, the radiation unit **11** has a good reflection coefficient when transmitting or receiving signals, and the **S12** value between the signal receiving end RX and the signal transmitting end TX of the radiation unit **11** can reach below -40 dB, which can effectively reduce the coupling effect between the first feed line **1331** and the second feed line **1332**, so that the first feed line **1331** and the second feed line **1332** have better isolation.

[0100] Referring to FIG. 18, FIG. 18 is an antenna gain curve diagram of the radiation unit **11** using FIG. 16 as a coupling layer. Curve **S181** represents a gain value of the signal transmitting end TX when the radiation unit **11** is used to transmit signals; curve **S182** represents a gain value of the signal receiving end RX when the radiation unit **11** is used to receive signals. As can be seen from FIG. 18, the radiation unit **11** of this example has a high antenna gain, which meets the antenna working requirements.

[0101] Referring to FIG. 19, FIG. 19 shows a schematic diagram of the structure of the coupling layer **131e** of a fifth specific example of the embodiment. As shown in FIG. 19, both ends of the first coupling slot **1311e** are formed with the port shapes, and the port shapes form an angle, the angle is oriented toward the first coupling slot **1311e**. Both ends of the second coupling slot **1312e** are formed with the port shapes, and the port shapes form an angle, the angle is facing away from the second coupling slot **1312e**.

[0102] Specifically, the first coupling slot **1311e** includes a first main slot **13110e** which is narrow and long, a first sub-slot **13111e** and a second sub-slot **13112e** which are connected to the first main slot **13110e** are respectively formed at both ends of the first main slot **13110e**, the first sub-slot **13111e** and the second sub-slot **13112e** are formed with an angle as a whole, and the angle is facing away from the first main slot **13110e**, the first sub-slot **13111e** and the second sub-slot **13112e** are symmetrically arranged along a center of the first main slot **13110e**, so both ends of the first coupling slot **1311e** are formed with the port shape. The second coupling slot **1312e** includes a second main slot **13120e** which is narrow and long, a third sub-slot **13121e** and a fourth sub-slot **13122e** that are connected to the second main slot **13120e** are respectively formed at both ends of the second main slot **13120e**, the third sub-slot **13121e** and the fourth sub-slot **13122e** are formed with an angle as a whole, and the angle is facing away from the second main slot **13120e**, the third sub-slot **13121e** and the fourth sub-slot **13122e** are symmetrically arranged along a center of the second main slot **13120e**, so both ends of the second coupling slot **1312e** are formed with the port shape. Of course, it is understood that, in some other embodiments, the first sub-slot **13111e** and the second sub-slot **13112e** may not be symmetrically arranged along the center of the first main slot **13110e**, and the third sub-slot **13121e** and the fourth sub-slot **13122e** may not be symmetrically arranged along the center of the second main slot **13120e**. An extension line of the first main slot **13110e** of the first coupling slot **1311e** and the second main slot **13120e** of the second coupling slot **1312e** are perpendicular to each other.

[0103] Referring to FIG. 20, FIG. 20 is a scattering parameter (S-parameter) curve diagram of the radiation unit **11** using FIG. 19 as a coupling layer. Curve **S191** represents a **S11** value of the signal receiving end RX when the radiation unit **11** is used to receive signals; curve **S192** represents a **S11** value of the signal transmitting end TX when the radiation unit **11** is used to transmit signals; curve **S193** represents a **S12** value between the signal receiving end RX and the signal transmitting end TX. As can be seen from FIG. 20, the radiation unit **11** has a good reflection coefficient when transmitting or receiving signals, and the **S12** value between the signal receiving end RX and the signal transmitting end TX of the radiation unit **11** can reach below -40 dB, which can effectively reduce the coupling effect between the first feed line **1331** and the second feed line **1332**, so that the first feed line **1331** and the second feed line **1332** have better isolation.

[0104] Referring to FIG. 21, FIG. 21 is an antenna gain curve diagram of the radiation unit **11** using FIG. 19 as a coupling layer. A gain value of the signal transmitting end TX when the radiation unit **11** is used to transmit signals is substantially coincident with a gain value of the signal receiving end RX when the radiation unit **11** is used to receive signals. As can be seen from FIG. 21, the radiation unit **11** of this example has a high antenna gain, which meets the antenna working requirements.

[0105] Referring to FIG. 22, FIG. 22 shows a schematic diagram of the structure of the coupling layer **131f** of a sixth specific example of the embodiment. As shown in FIG. 22, both ends of the first coupling slot **1311f** are formed with the port shapes, and the port shapes form an angle, the angle is facing away from the first coupling slot **1311f**. Both ends of the second coupling slot **1312f** do not form the port shapes with an angle.

[0106] Specifically, the first coupling slot **1311f** includes a first main slot **13110f** which is narrow and long, a first sub-slot **13111f** and a second sub-slot **13112f** which are connected to the first main slot **13110f** are respectively formed at both ends of the first main slot **13110f**, the first sub-slot **13111f** and the second sub-slot **13112f** are formed with an angle as a whole, and the angle is facing away from the first main slot **13110f**, the first sub-slot **13111f** and the second sub-slot **13112f** are symmetrically arranged along a center of the first main slot **13110f**, so both ends of the first coupling slot **1311f** are formed with the port shape. The second coupling slot **1312f** includes a second main slot **13120f** which is narrow and long, a third sub-slot **13121f** and a fourth sub-slot **13122f** that are connected to the second main slot **13120f** are respectively formed at both ends of the second main slot **13120f**, the third sub-slot **13121f** and the fourth sub-slot **13122f** are both vertically connected to the second main slot **13120f**, so that the second coupling slot **1312f** is H-shaped as a whole. An extension line of the first main slot **13110f** of the first coupling slot **1311f** and the second main slot **13120f** of the second coupling slot **1312f** are perpendicular to each other.

[0107] Referring to FIG. 23, FIG. 23 is a scattering parameter (S-parameter) curve diagram of the radiation unit **11** using FIG. 22 as a coupling layer. Curve **S231** represents a **S11** value of the signal receiving end RX when the radiation unit **11** is used to receive signals; curve **S232** represents a **S11** value of the signal transmitting end TX when the radiation unit **11** is used to transmit signals; curve **S233** represents a **S12** value between the signal receiving end RX and the signal transmitting end TX. As can be seen from FIG. 23, the radiation unit **11** has a good reflection coefficient when transmitting or receiving signals, and the **S12** value between the signal receiving end RX and the signal transmitting end TX of the radiation unit **11** can reach below -50 dB, which can effectively reduce the coupling effect between the first feed line **1331** and the second feed line **1332**, so that the first feed line **1331** and the second feed line **1332** have better isolation.

[0108] Referring to FIG. 24, FIG. 24 is an antenna gain curve diagram of the radiation unit **11** using FIG. 22 as a coupling layer. A gain value of the signal transmitting end TX when the radiation unit **11** is used to transmit signals is substantially coincident with a gain value of the signal receiving end RX when the radiation unit **11** is used to receive signals. As can be seen from FIG. 24, the radiation unit **11** of this example has a high antenna gain, which meets the antenna working requirements.

[0109] Referring to FIG. 25, FIG. 25 shows a schematic diagram of the structure of the coupling layer **131g** of a seventh specific example of the embodiment. As shown in FIG. 25, both ends of the first coupling slot **1311g** do not form the port shapes with an angle. Both ends of the second coupling slot **1312g** are formed with the port shapes, and the port shapes form an angle, the angle is facing away from the second coupling slot **1312g**.

[0110] Specifically, the first coupling slot **1311g** includes a first main slot **13110g** which is narrow and long, a first sub-slot **13111g** and a second sub-slot **13112g** which are connected to the first main slot **13110g** are respectively formed at both ends of the first main slot **13110g**, the first sub-slot **13111g** and the second sub-slot **13112g** are both vertically connected to the first main slot **13110g**, so that the first coupling slot **1311g** is H-shaped as a whole. The second coupling slot **1312g** includes a second main slot **13120g** which is narrow and long, a third sub-slot **13121g** and a fourth sub-slot **13122g** that are connected to the second main slot **13120g** are respectively formed at both ends of the second main slot **13120g**, the third sub-slot **13121g** and the fourth sub-slot **13122g** are formed with an angle as a whole, and the angle is facing away from the second main slot **13120g**, the first sub-slot **13111g** and the second sub-slot **13112g** are symmetrically arranged along a center of the second main slot **13120g**, so both ends of the second main slot **13120g** are formed with the port shape. An extension line of the first main slot **13110g** of the first coupling slot **1311g** and the second main slot **13120g** of the second coupling slot **1312g** are perpendicular to each other.

[0111] Referring to FIG. 26, FIG. 26 is a scattering parameter (S-parameter) curve diagram of the radiation unit **11** using FIG. 25 as a coupling layer. Curve **S261** represents a **S11** value of the signal receiving end RX when the radiation unit **11** is used to receive signals; curve **S262** represents a **S11** value of the signal transmitting end TX when the radiation unit **11** is used to transmit signals; curve **S263** represents a **S12** value between the signal receiving end RX and the signal transmitting end TX. As can be seen from FIG. 26, the radiation unit **11** has a good reflection coefficient when transmitting or receiving signals, and the **S12** value between the signal receiving end RX and the signal transmitting end TX of the radiation unit **11** can reach below -50 dB, which can effectively reduce the coupling effect between the first feed line **1331** and the second feed line **1332**, so that the first feed line **1331** and the second feed line **1332** have better isolation.

[0112] Referring to FIG. 27, FIG. 27 is an antenna gain curve diagram of the radiation unit **11** using FIG. 25 as a coupling layer. A gain value of the signal transmitting end TX when the radiation unit **11** is used to transmit signals is substantially coincident with a gain value of the signal receiving end RX when the radiation unit **11** is used to receive signals. As can be seen from FIG. 27, the radiation unit **11** of this example has a high antenna gain, which meets the antenna working requirements.

[0113] Referring to FIG. 28, FIG. 28 shows a schematic diagram of the structure of the coupling layer **131h** of a seventh specific example of the embodiment. As shown in FIG. 28, one end of the first coupling slot **1311h** is formed with the port shape, and the port shape form an angle, the angle is orientated towards the first coupling slot **1311h**. Both ends of the second coupling slot **1312h** do not form the port shapes with an angle.

[0114] Specifically, the first coupling slot **1311h** includes a first main slot **13110h** which is narrow and long, a first sub-slot **13111h** which is connected to the first main slot **13110h** is formed at one end of the first main slot **13110h**, the first sub-slot **13111h** is formed with an angle as a whole, and the angle is orientated towards the first main slot **13110h**, so the first coupling slot **1311h** is the shape of a unidirectional arrow. The first sub-slot **13111h** closes to the second coupling slot **1312h**. The second coupling slot **1312g** includes a second main slot **13120h** which is narrow and long. An extension line of the first main slot **13110h** of the first coupling slot **1311h** and the second main slot **13120h** of the second coupling slot **1312h** are perpendicular to each other.

[0115] Referring to FIG. 29, FIG. 29 is a scattering parameter (S-parameter) curve diagram of the radiation unit **11** using FIG. 28 as a coupling layer. Curve **S291** represents a **S11** value of the signal

receiving end RX when the radiation unit **11** is used to receive signals; curve **S292** represents a **S11** value of the signal transmitting end TX when the radiation unit **11** is used to transmit signals; curve **S293** represents a **S12** value between the signal receiving end RX and the signal transmitting end TX. As can be seen from FIG. **29**, the radiation unit **11** has a good reflection coefficient when transmitting or receiving signals, and the **S12** value between the signal receiving end RX and the signal transmitting end TX of the radiation unit **11** can reach below -50 dB, which can effectively reduce the coupling effect between the first feed line **1331** and the second feed line **1332**, so that the first feed line **1331** and the second feed line **1332** have better isolation.

[0116] Referring to FIG. **30**, FIG. **30** is an antenna gain curve diagram of the radiation unit **11** using FIG. **28** as a coupling layer. Curve **S301** represents a gain value of the signal transmitting end TX when the radiation unit **11** is used to transmit signals; curve **S302** represents a gain value of the signal receiving end RX when the radiation unit **11** is used to receive signals. As can be seen from FIG. **30**, the radiation unit **11** of this example has a high antenna gain, which meets the antenna working requirements.

[0117] Of course, the coupling layer **131** of the embodiment of the present application is not limited to the above specific examples. For example, the port shape formed with an angle can also be formed only at one end of the second coupling slot **1312**, and the angle can be toward the second coupling slot **1312**, and can also be facing away from the second coupling slot **1312**.

[0118] Referring to FIG. **31**, FIG. **31** shows a schematic diagram of the structure of the coupling layer **131i** of a ninth specific example of the embodiment. As shown in FIG. **31**, both ends of each of the first coupling slot **1311i** and the second coupling slot **1312i** are formed with the port shapes, and the port shape does not form an angle.

[0119] Specifically, the first coupling slot **1311i** includes a first main slot **13110i** which is narrow and long, a first sub-slot **13111i** and a second sub-slot **13112i** which are connected to the first main slot **13110i** are respectively formed at both ends of the first main slot **13110i**, the first sub-slot **13111i** and the second sub-slot **13112i** are both vertically connected to the first main slot **13110i**, the first sub-slot **13111i** and the second sub-slot **13112i** are symmetrically arranged along a center of the first main slot **13110i**. The second coupling slot **1312i** includes a second main slot **13120i** which is narrow and long, a third sub-slot **13121i** and a fourth sub-slot **13122i** that are connected to the second main slot **13120i** are respectively formed at both ends of the second main slot **13120i**, the third sub-slot **13121i** and the fourth sub-slot **13122i** are both vertically connected to the second main slot **13120i**, the third sub-slot **13121i** and the fourth sub-slot **13122i** are symmetrically arranged along a center of the second main slot **13120i**. So each of the first coupling slot **1311i** and the second coupling slot **1312i** is H-shaped as a whole.

[0120] An extension line of the first main slot **13110i** of the first coupling slot **1311i** and the second main slot **13120i** of the second coupling slot **1312i** are perpendicular to each other. An intersection of the extension line of the first main slot **13110i** of the first coupling slot **1311i** and the second main slot **13120i** of the second coupling slot **1312i** may be a midpoint of the second main slot **13120i**. Of course, it is understood that, in some other embodiments, the first sub-slot **13111i** and the second sub-slot **13112i** may not be symmetrically arranged along the center of the first main slot **13110i**, the third sub-slot **13121i** and the fourth sub-slot **13122i** may not be symmetrically arranged along the center of the second main slot **13120i**.

[0121] Referring to FIG. **32**, FIG. **32** is a schematic diagram of current paths of the coupling layer **131i** in the example of FIG. **31**. According to FIG. **32**, the current path and direction of the current on the coupling layer **131a** are changed by the H-shaped of the first coupling slot **1311i** and the second coupling slot **1312i**, the current is concentrated near the port shape formed by the first coupling slot **1311i** and the second coupling slot **1312i**, thereby reducing the coupling interference between the first feed line **1331i** and the second feed line **1332i**, improving the isolation between the first feed line **1331i** and the second feed line **1332i**, and thus reducing the interference effect of the antenna array **10**. However, compared with the current path schematic diagram of the coupling

layer **131a** of the first specific example shown in FIG. 6 and the current path schematic diagram of the coupling layer **131b** shown in FIG. 10, the port shaped with an angle formed in the coupling layer **131a** in the example of FIG. 5 and the coupling layer **131b** in the example of FIG. 9 has a greater degree of change in the current path and direction. Therefore, compared with the example of FIG. 31, the coupling layer **131i** can further reduce the coupling interference between the first feed line **1331** and the second feed line **1332**, further improve the isolation between the first feed line **1331** and the second feed line **1332**, and further reduce the interference effect of the antenna array **10**.

[0122] Referring to FIG. 33, FIG. 33 is a scattering parameter (S-parameter) curve diagram of the radiation unit **11** using FIG. 31 as a coupling layer. Curve **S331** represents a **S11** value of the signal receiving end RX when the radiation unit **11** is used to receive signals; curve **S332** represents a **S11** value of the signal transmitting end TX when the radiation unit **11** is used to transmit signals; curve **S333** represents a **S12** value between the signal receiving end RX and the signal transmitting end TX. As can be seen from FIG. 33, the reflection coefficient of the radiation unit **11** when transmitting signals is approximately -15 dB, the reflection coefficient of the radiation unit **11** when receiving signals is approximately -14 dB, and the lowest value of the **S12** value between the signal receiving end RX and the signal transmitting end TX of the radiation unit **11** is approximately -38 dB. Compared with the S-parameter curves of the radiation unit **11** in the above-mentioned specific examples, although the radiation unit **11** in this example can reduce the coupling effect between the first feed line **1331** and the second feed line **1332**, so that there is a certain isolation between the first feed line **1331** and the second feed line **1332**, the effect is slightly worse than that of the above-mentioned examples.

[0123] In summary, the antenna array **10** provided by this application includes the first radiation module **110** and the feeding module **130**. The first radiation module **110** includes the first radiation layer **111** and the first dielectric layer **112** arranged in stack, and the plurality of first radiators **1111** are provided on the first radiation layer **111**. In this way, the first dielectric layer **112** can concentrate the antenna beam of each first radiator **1111** to improve the antenna gain of the antenna array **10**. The feeding module **130** includes the feeding layer **133** and the coupling layer **131**, the feeding layer **133** is provided with the first feed lines **1331** and the second feed lines **1332**, the first feed lines **1331** and the second feed lines **1351** are arranged in one-to-one correspondence to form the plurality of feed line units, and each feed line unit corresponds to the first radiator **1111** in one-to-one correspondence. In this way, the first feed lines **1331** and the second feed lines **1351** can generate different polarized waves respectively, so that the first radiators **1111** can receive and transmit signals of two different polarized states at the same time, which can improve a system capacity of the antenna array **10**, reduce interference from the antenna array **10**, enhance signal quality, and improve coverage of the antenna array **10**. The coupling layer **131** is provided with the first coupling slots **1311** and the second coupling slots **1312**, the first coupling slots **1311** and the second coupling slots **1312** are arranged in one-to-one correspondence to form a plurality of coupling units. The first coupling slots **1311** and the second coupling slots **1312** of the coupling units are arranged at intervals from each other. In this way, when the coupling layer **131** couples the current signal flowing through the feeding units to the first radiator **1111**, the coupling layer **131** changes the current path and direction of the coupling layer **131** through the first coupling slot **1311** and the second coupling slot **1312**, thereby reducing the coupling interference between the first feed line **1331** and the second feed line **1332**, thereby reducing the interference effect of the antenna array **10**.

[0124] Further, when the extension line of the first coupling slot **1311** is perpendicular to the extension line of the second coupling slot **1312**, the ends of the first coupling slot **1311** and the second coupling slot **1312** form the port shape, and the port shape is formed with an angle, the port shape formed with the angle can change the current path and direction on the coupling layer **131**, and the current is concentrated near the port shape, so the coupling interference between the first

feed line **1331** and the second feed line **1332** can be further reduced, and the isolation between the first feed line **1331** and the second feed line **1332** can be effectively improved, thereby effectively reducing the interference effect of the antenna array **10**.

Embodiment II

[0125] Please refer to FIGS. **34** and **35**, embodiment II of the present application further provides another structure of the radiation unit **11a**. The structure of the radiation unit **11a** provided in the embodiment II is substantially the same as the structure of the radiation unit **11** provided in the embodiment I, the difference is that at least one radiation module in the embodiment II includes two radiation modules. And the radiation layer of one of the radiation modules is disposed close to the dielectric layer of the other one of the radiation modules. The radiators of the two radiation layers are arranged in one-to-one correspondence, and the two corresponding radiators receive the feed signal provided by the same feeding unit.

[0126] For instance, the antenna array **10** further includes a second radiation module **140**. The second radiation module **140** includes a second radiation layer **141** and a second dielectric layer **142** arranged in stack. The second radiation layer **141** includes a plurality of second radiators **1411**. The first radiation layer **111** of the first radiation module **110** is disposed close to the second dielectric layer **142** of the second radiation module **140**. And the plurality of second radiators **1411** and the plurality of first radiators **1111** are arranged in one-to-one correspondence. In this way, the first radiators **1111** and the second radiators **1411** corresponding to each other and can receive the feed signal coupled to the feed line unit corresponding to the first radiator **1111**. That is to say, after the feed line unit corresponding to the first radiator **1111** couples energy to the first radiator **1111**, the first radiator **1111** continues to couple energy to the second radiator **1411** to realize signal transmission or reception of the antenna array **10**. In the embodiment, the first radiation layer **111** can be a printed circuit board, the second radiator **1411** can be a substantial circular metal sheet or metal coating formed on the first radiation layer **111**.

[0127] Further, the second dielectric layer **142** is also provided with a plurality of cavities **1421**. Each cavity **1421** penetrates the second dielectric layer **142**. A diameter of the cavity **1421** may be equal to the diameter of the corresponding protection cavity **121**, an edge of the cavity **1421** is aligned with an edge of the protection cavity **121**. And each cavity **1421** is provided in one-to-one correspondence with the two radiators on both sides (i.e., the first radiator **1111** and the second radiator **1411**). Specifically, a line formed by centers of the cavity **1421**, the first radiator **1111**, and the second radiator **1411** is parallel to the Z-axis. Moreover, a projected area of the second radiator **1411** in the Z-axis direction of the antenna array **10** is larger than the projected area of the first radiator **1111** in the Z-axis direction of the antenna array **10**. Thus, in the embodiment, the antenna height is further increased through the first dielectric layer **112** and the second dielectric layer **142** to increase the antenna gain; the second radiator **1411** is provided to cover the first radiator **1111** so that the energy that coupled to the second radiator **1411** by the first radiator **1111** is more concentrated, thereby increasing the directivity of the energy beam of the antenna array **10**.

[0128] It can be understood that the second dielectric layer **142** can also be made of plastic or ceramic materials.

[0129] It can be understood that this application does not limit the specific shapes of the first radiator **1111** and the second radiator **1411**. For example, in other embodiments, the shape of the first radiator **1111** may also be a rectangle. In other embodiments, the shapes of the first radiator **1111** and the second radiator **1411** may also be other polygonal or irregular shapes, and the shapes of the first radiator **1111** and the second radiator **1411** may be the same or different. It is only necessary that the projected area of the first radiator **1111** in the Z-axis direction of the antenna array **10** can cover the projected area of the coupling slot **1311**, and the projected area of the second radiator **1411** in the Z-axis direction of the antenna array **10** can cover the first radiator **1111**.

[0130] It can be understood that the working principle of the antenna array **10** provided in the embodiment II is substantially the same as that of the antenna array provided in the embodiment I.

The difference is that after the feed line unit couples energy to the first radiator **1111** through the coupling slot **1311** of the coupling layer **131**, the first radiator **1111** continues to couple energy to the second radiator **1411** to transmit left-hand polarized waves or right-hand polarized waves through the second radiator **1411**, thereby realizing communication between the antenna array **10** and the low-orbit satellite.

[0131] Referring to FIG. **36**, one embodiment of the present application also provides a radio frequency (RF) module **100**. The radio frequency module **100** includes a transmitter **150** and a receiver **160**. It can be understood that the transmitter **150** and the receiver **160** form a signal transceiver electrically connected to the antenna array **10** (not shown in FIG. **26**, which can be referred to the antenna array **10** described above) and for providing or receiving signals to the first feed line **1331** and the second feed line **1332** in the antenna array **10**.

[0132] The antenna array **10** is not limited to the antenna array **10** mentioned in the embodiment I, and may also be the antenna array composed of the radiation units provided in the embodiment II. In this way, the radio frequency module **100**, by providing the antenna array **10**, provided in this application can realize communication between the device installed with the radio frequency module **100** and low-orbit satellites.

[0133] Referring to FIG. **37**, one embodiment of the present application also provides an electronic device **200**, including a processor (such as CPU) **210**, the antenna array **10**, and the radio frequency module **100** shown in FIG. **36**. The processor **210** is used to modulate the communication signal that needs to be radiated outward and transmit it to the transmitter **150** of the radio frequency module **100**, the transmitter **150** receives the modulated communication signal to generate the feed signal, and transmits the feed signal to the first feed line **1331** and the second feed line **1332** of the antenna array **10**. The processor **210** is also used to receive the external signal received by the receiver **160** of the radio frequency module **100** through the first feed line **1331** and the second feed line **1332** of the antenna array **10**, and demodulate the external signal. By providing the radio frequency module **100**, the electronic device **200** can communicate with low-orbit satellites. The electronic device **200** may be a ground station, a mobile vehicle, or other electronic device that needs to communicate with a low-orbit satellite.

[0134] The embodiments shown and described above are only examples. Even though numerous characteristics and advantages of the present technology have been set forth in the foregoing description, together with details of the structure and function of the present disclosure, the disclosure is illustrative only, and changes may be made in the detail, including in matters of shape, size and arrangement of the parts within the principles of the present disclosure, up to and including the full extent established by the broad general meaning of the terms used in the claims.

Claims

1. An antenna array comprising: at least one radiation module comprising a radiation layer and a dielectric layer arranged in stack, the radiation layer comprising a plurality of radiators, each of the plurality of radiators configured to radiate signals, the dielectric layer being made of non-conductive material; and a feeding module arranged on a side of the radiation module close to the dielectric layer, the feeding module comprising a feeding layer and a coupling layer arranged in stack, the feeding layer comprising a plurality of first feed lines and a plurality of second feed lines, the plurality of first feed lines and the plurality of second feed lines being one-to-one correspondingly arranged to form a plurality of feeding units, the plurality of feeding units being one-to-one corresponding to the plurality of radiators and feed current to the plurality of radiators, wherein the coupling layer is arranged on a side of the feeding layer close to the first radiation module, the coupling layer is configured to couple the current flowing through the feeding units to the radiation module; and the coupling layer is provided with a plurality of first coupling slots and a plurality of second coupling slots, the plurality of first coupling slots and the plurality of second

coupling slots are one-to-one correspondingly arranged to form a plurality of coupling units; each of the plurality of first coupling slots is spaced apart from each of the plurality of second coupling slots.

2. The antenna array of claim 1, wherein the plurality of first coupling slots and the plurality of second coupling slots extend in different directions.

3. The antenna array of claim 1, wherein in each of the plurality of coupling units, a shape of the first coupling slot is different from a shape of the second coupling slot.

4. The antenna array of claim 2, wherein in each of the plurality of coupling units, at least one end of at least one of the first coupling slot and the second coupling slot is formed with a port shape.

5. The antenna array of claim 1, wherein the plurality of coupling units and the plurality of feeding units of the coupling layer are one-to-one correspondingly arranged; each of the plurality of first coupling slots is arranged correspondingly to a first feed line, each of the plurality of second coupling slots is arranged corresponding to a corresponding second feed line.

6. The antenna array of claim 1, wherein a length of each of the plurality of first coupling lines is same to a length of each of the plurality of second coupling lines.

7. The antenna array of claim 1, wherein the antenna array comprises two radiation modules, the radiation layer of one of the two radiation modules is arranged close to the dielectric layer of the other of the two radiation modules, the plurality of radiators of the two radiation layers are one-to-one correspondingly arranged, and the two corresponding radiators are fed by a same feeding unit.

8. The antenna array of claim 1, wherein the feeding module further comprises a first cavity layer, a second cavity layer, and a ground layer, the first cavity layer is disposed between the feeding layer and the coupling layer.

9. The antenna array of claim 1, wherein the feeding module further comprises a plurality of phase couplers, the plurality of phase couplers are arranged on the feeding layer.

10. The antenna array of claim 9, wherein the plurality of phase couplers are further connected to transmitters and receivers.

11. A ground station comprising an antenna array, the antenna array comprising: at least one radiation module comprising a radiation layer and a dielectric layer arranged in stack, the radiation layer comprising a plurality of radiators, each of the plurality of radiators configured to radiate signals, the dielectric layer being made of non-conductive material; and a feeding module arranged on a side of the radiation module close to the dielectric layer, the feeding module comprising a feeding layer and a coupling layer arranged in stack, the feeding layer comprising a plurality of first feed lines and a plurality of second feed lines, the plurality of first feed lines and the plurality of second feed lines being one-to-one correspondingly arranged to form a plurality of feeding units, the plurality of feeding units being one-to-one corresponding to the plurality of radiators and feed current to the plurality of radiators, wherein the coupling layer is arranged on a side of the feeding layer close to the first radiation module, the coupling layer is configured to couple the current flowing through the feeding units to the radiation module; and the coupling layer is provided with a plurality of first coupling slots and a plurality of second coupling slots, the plurality of first coupling slots and the plurality of second coupling slots are one-to-one correspondingly arranged to form a plurality of coupling units; each of the plurality of first coupling slots is spaced apart from each of the plurality of second coupling slots.

12. The ground station of claim 11, wherein the plurality of first coupling slots and the plurality of second coupling slots extend in different directions.

13. The ground station of claim 11, wherein in each of the plurality of coupling units, a shape of the first coupling slot is different from a shape of the second coupling slot.

14. The ground station of claim 12, wherein in each of the plurality of coupling units, at least one end of at least one of the first coupling slot and the second coupling slot is formed with a port shape.

15. The ground station of claim 11, wherein the plurality of coupling units and the plurality of

feeding units of the coupling layer are one-to-one correspondingly arranged; each of the plurality of first coupling slots is arranged correspondingly to a first feed line, each of the plurality of second coupling slots is arranged corresponding to a corresponding second feed line.

16. The ground station of claim 11, wherein a length of each of the plurality of first coupling lines is same to a length of each of the plurality of second coupling lines.

17. The ground station of claim 11, wherein the antenna array comprises two radiation modules, the radiation layer of one of the two radiation modules is arranged close to the dielectric layer of the other of the two radiation modules, the plurality of radiators of the two radiation layers are one-to-one correspondingly arranged, and the two corresponding radiators are fed by a same feeding unit.

18. The ground station of claim 11, wherein the feeding module further comprises a first cavity layer, a second cavity layer, and a ground layer, the first cavity layer is disposed between the feeding layer and the coupling layer.

19. The ground station of claim 11, wherein the feeding module further comprises a plurality of phase couplers, the plurality of phase couplers are arranged on the feeding layer.

20. The ground station of claim 19, wherein the plurality of phase couplers are further connected to transmitters and receivers.
